WorldWright Current-Repo Generate Diagnostics - Branch / PR 67 Context

This investigation was done against the current checked-out repo state, not an old snapshot.

Important branch context:

- Current local branch: work.
- The branch history shows PR #67 was merged, then several follow-up commits reverted or removed parts of PR #67.
- This diagnostic reflects the current code after PR #67 plus later removals/reverts, not the original PR #67 merge commit by itself.

Relevant recent history:

```
work
d82e1de Restore worldCrust export formatting
33761e0 Remove PR #67 crust material authority tests
c9ed6b2 Remove PR #67 material tiny cleanup
2d2839f Remove PR #67 material terrain subpasses
8dd1a1d Remove PR #67 crust material helpers
6526b8c Remove PR #67 crust material blueprint note
9ac936b Revert PR #67 crust material terrain route
65e474a Merge pull request #67 from smchammer3-cyber/crust-material-fields-feature-relief
```

The current Generate pipeline order is:

```
seed continent skeleton fields
-> apply skeleton base elevation
-> recompute
-> apply generated world quality pass
-> recompute
-> seed continent skeleton fields
-> seed crust fields
-> apply crust terrain influence
-> apply ocean bathymetry smoothing
-> recompute
-> seed continent skeleton fields
-> seed crust fields
```

The current stage diagnostics replay isolates these Generate stages:

```
RAW_GENERATOR
CONTINENT_FIELDS
SKELETON_ELEVATION
FIRST_RECOMPUTE
QUALITY_PASS
CRUST_FIELDS
CRUST_PROVINCE_DELTA
CRUST_COAST_BREAKUP
```

CRUST_COHERENCE
CRUST_SKELETON_OBEDIENCE
CRUST_TINY_ISLAND_CLEANUP
OCEAN_BATHYMETRY_SMOOTHING
FINAL_RECOMPUTE
FINAL_CONTINENT_RESEED
FINAL_CRUST_RESEED

Diagnostic sample used:

width = 128
height = 64
seed = "67"
continentCount = 4

Short Answers

1. Which Generate stage first causes plate/skeleton/province imprint to jump?

Plate imprint first jumps at SKELETON_ELEVATION.

RAW_GENERATOR plate imprint: 0.882x
CONTINENT_FIELDS plate imprint: 0.882x
SKELETON_ELEVATION plate imprint: 1.209x

Skeleton classification imprint appears as soon as CONTINENT_FIELDS are seeded:

CONTINENT_FIELDS skeleton imprint: 1.362x

Province imprint first appears at CRUST_FIELDS:

CRUST_FIELDS province imprint: 1.503x

Then province imprint becomes more terrain-visible at CRUST_PROVINCE_DELTA:

CRUST_PROVINCE_DELTA province imprint: 1.769x

Final answer:

First plate height imprint jump: SKELETON_ELEVATION
First skeleton identity imprint: CONTINENT_FIELDS
First province identity imprint: CRUST_FIELDS

First province terrain imprint:     CRUST_PROVINCE_DELTA

2. Is skeleton elevation over-amplifying plate-shaped structure before crust runs?

Somewhat, yes.

SKELETON_ELEVATION is the first stage that turns hidden continent/skeleton structure into visible height structure.

In seed 67:

Plate imprint:    0.882x -> 1.209x
Land fraction:    31.13% -> 29.06%
Land bodies:      44 -> 35
Topology flip:    5.82%
Land relief:      0.0338 -> 0.0341

That means skeleton elevation is not just harmless guidance. It significantly changes terrain and sea-level topology.

However, it is not totally uncontrolled. The current code has safeguards:

- land protection
- weak-ocean uplift reduction
- invalid-fragment handling
- caused-island exceptions
- seam damping

Conclusion:

Skeleton elevation is restrained, but yes, it is the first stage that visibly amplifies hidden plate/skeleton structure before crust runs.

3. Does crust delta create too many land bodies/fragments?

It creates extra land bodies, but not mainly medium fragments.

Before crust delta:

CRUST_FIELDS
land bodies:        34
medium fragments:   4
medium share:       7.8%

After crust delta:

CRUST_PROVINCE_DELTA
land bodies: 41
medium fragments: 4
medium share: 7.72%

So crust delta adds:

+7 land bodies
+0 medium fragments

Conclusion:

Crust delta does increase total land-body count, but in this current code sample it does not explode medium fragments.

4. Is crust cohere mostly repairing damage caused by crust delta?

Partly, but no - it is not the main repair pass.

Sequence after crust delta:

CRUST_PROVINCE_DELTA:
land bodies: 41
medium fragments: 4

CRUST_COAST_BREAKUP:
land bodies: 40
medium fragments: 4

CRUST_COHERENCE:
land bodies: 38
medium fragments: 4

CRUST_SKELETON_OBEDIENCE:
land bodies: 36
medium fragments: 2

CRUST_TINY_ISLAND_CLEANUP:
land bodies: 29
medium fragments: 2

Crust cohere repairs some damage:

land bodies: 40 -> 38

But the bigger repairs come later:

CRUST_SKELETON_OBEDIENCE:
medium fragments: 4 -> 2

CRUST_TINY_ISLAND_CLEANUP:
land bodies: 36 -> 29

Conclusion:

Crust cohere helps, but the main repair burden is carried by crust skeleton obedience and tiny island cleanup.

5. Is crust skeleton double-applying skeleton authority?

Yes, but currently in a restrained way.

There are two skeleton terrain authority stages:

1. Early skeleton elevation:
   applySkeletonBaseElevation

2. Late crust skeleton obedience:
   applyContinentSkeletonTerrainObedience

The current code explicitly says the late crust skeleton pass is intended to be restrained and should not behave like a second continent generator.

But in seed 67 it still has a real effect:

```
Land fraction:      29.16% -> 30.69%
Land bodies:        38 -> 36
Medium fragments:   4 -> 2
Topology flip:      1.65%
Plate imprint:      1.382x -> 1.430x
Province imprint:   1.785x -> 1.839x
Skeleton imprint:   1.374x -> 1.386x
```

Conclusion:

Yes. Crust skeleton is a second skeleton-authority pass. It is restrained and useful for repair, but it still changes terrain and increases imprint.

6. Are crust fields derived from height/oceanDepthClass in a way that creates feedback?

Yes. This is one of the clearest current feedback risks.

Current crust field seeding reads current terrain and ocean classification:

```
height
aboveSea
terrainBuoyancy
oceanDepthClass
plateType
boundaryType
upliftRate
volcanicActivity
surfaceAge
```

Then it writes:

```
crustThickness
crustAge
crustProvince
```

Then the next terrain pass reads those crust fields and modifies height:

```
crustThickness / crustAge / crustProvince
-> crust province terrain delta
-> changed height
```

So the loop is:

```
height / oceanDepthClass
-> crust fields
-> crust province
-> height delta
-> final terrain
-> final crust reseed
```

Conclusion:

Yes. Crust fields are partly height-derived, then crust fields modify height. That creates a height -> crust -> height feedback loop.

7. Which stage helps land relief, and which stage hurts it?

Using landHeightStdDev as the relief metric:

RAW_GENERATOR: 0.0338
SKELETON_ELEVATION: 0.0341
QUALITY_PASS: 0.0329
CRUST_FIELDS: 0.0329
CRUST_PROVINCE_DELTA: 0.0313
CRUST_COAST_BREAKUP: 0.0314
CRUST_COHERENCE: 0.0314
CRUST_SKELETON_OBEDIENCE: 0.0324
FINAL: 0.0324

Stages that help relief:

SKELETON_ELEVATION
CRUST_COAST_BREAKUP, slightly
CRUST_SKELETON_OBEDIENCE

Stages that hurt relief:

QUALITY_PASS
CRUST_PROVINCE_DELTA

Conclusion:

Best relief helper: CRUST_SKELETON_OBEDIENCE
Secondary helper: SKELETON_ELEVATION
Biggest relief hurt: CRUST_PROVINCE_DELTA
Unexpected relief hurt: QUALITY_PASS

8. Which pass most increases export risk?

There are two kinds of export risk:

A. terrain/export height risk
B. metadata/debug/color/province export risk

Biggest terrain export risk: CRUST_PROVINCE_DELTA.

It changes height based on crust province and increases visible imprint:

Plate imprint: 1.227x -> 1.344x
Province imprint: 1.503x -> 1.769x
Province leak: 13.53% -> 18.70%

Land bodies: 34 -> 41

Biggest metadata/debug/export classification risk: CRUST_FIELDS.

This stage does not change height, but it creates crust province labels that immediately have strong spatial imprint:

Province imprint appears at: 1.503x
Province leak jumps to: 13.53%
Ocean leak jumps to: 13.31%
Land leak jumps to: 9.44%

Final export metadata risk: FINAL_CRUST_RESEED.

This does not change terrain, but it changes/re-syncs final crust/province identity after terrain has settled.

In seed 67:

Province imprint: 1.848x -> 1.870x
Province leak: 20.30% -> 20.76%
Ocean leak: 15.06% -> 15.92%

Conclusion:

Most terrain export risk: CRUST_PROVINCE_DELTA
Most classification/export risk: CRUST_FIELDS
Final export label risk: FINAL_CRUST_RESEED
Most visible late terrain risk: CRUST_SKELETON_OBEDIENCE

Full Stage Table

| Stage | Land | Bodies | Medium | Relief | Plate | Province | Skeleton | Flip |
|---|---|---|---|---|---|---|---|---|
| RAW_GENERATOR | 31.13% | 44 | 9 | 0.0338 | 0.882x | - | - | - |
| CONTINENT_FIELDS | 31.13% | 44 | 9 | 0.0338 | 0.882x | - | 1.362x | 0% |
| SKELETON_ELEVATION | 29.06% | 35 | 3 | 0.0341 | 1.209x | - | 1.362x | 5.82% |
| FIRST_RECOMPUTE | 29.06% | 35 | 3 | 0.0341 | 1.209x | - | 1.362x | 0% |
| QUALITY_PASS | 28.97% | 34 | 4 | 0.0329 | 1.227x | - | 1.363x | 1.22% |
| CRUST_FIELDS | 28.97% | 34 | 4 | 0.0329 | 1.227x | 1.503x | 1.348x | 0% |
| CRUST_PROVINCE_DELTA | 29.41% | 41 | 4 | 0.0313 | 1.344x | 1.769x | 1.367x | 0.49% |
| CRUST_COAST_BREAKUP | 29.28% | 40 | 4 | 0.0314 | 1.377x | 1.784x | 1.374x | 0.24% |
| CRUST_COHERENCE | 29.16% | 38 | 4 | 0.0314 | 1.382x | 1.785x | 1.374x | 0.24% |
| CRUST_SKELETON_OBEDIENCE | 30.69% | 36 | 2 | 0.0324 | 1.430x | 1.839x | 1.386x | 1.65% |
| CRUST_TINY_ISLAND_CLEANUP | 30.48% | 29 | 2 | 0.0324 | 1.399x | 1.846x | 1.378x | 0.21% |
| OCEAN_BATHYMETRY_SMOOTHING | 30.48% | 29 | 2 | 0.0324 | 1.403x | 1.848x | 1.376x | 0% |
| FINAL_RECOMPUTE | 30.48% | 29 | 2 | 0.0324 | 1.403x | 1.848x | 1.376x | 0% |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| FINAL_CONTINENT_RESEED | 30.48% | 29 | 2 | 0.0324 | 1.403x | 1.848x | 1.457x | 0% |
| FINAL_CRUST_RESEED | 30.48% | 29 | 2 | 0.0324 | 1.403x | 1.870x | 1.457x | 0% |

Main Current-Repo Issues

Issue A - Skeleton elevation is the first major visible cause imprint

The raw generator has lower plate imprint:

0.882x

After skeleton elevation:

1.209x

This means skeleton elevation is the first pass that visibly increases plate-shaped structure.

Issue B - Crust fields are height-derived, so they can echo existing terrain

Crust fields are seeded from current height and ocean classification.

Risk:

terrain already has shape
-> crust fields classify from that shape
-> crust terrain pass reinforces that shape

This creates feedback.

Issue C - Province labels create export-visible seams before terrain is changed

CRUST_FIELDS does not change height, but province imprint immediately appears:

1.503x

This means province labels are already aligned with visible terrain/height structure.

That is risky for:

debug overlays
material exports
province exports
final color authority

height export explanations

Issue D - Crust province delta is the main terrain imprint amplifier

CRUST_PROVINCE_DELTA makes province identity visible in terrain:

Province imprint: 1.503x -> 1.769x
Land bodies:      34 -> 41

This is the main pass to inspect if hidden province structure is leaking into export height.

Issue E - Crust skeleton is useful but still double-applies skeleton authority

The late crust skeleton pass improves topology:

Medium fragments: 4 -> 2

But it also increases imprint:

Plate imprint:    1.382x -> 1.430x
Province imprint: 1.785x -> 1.839x

So it is useful but risky.

Issue F - Quality pass may not be helping land relief enough

The quality pass is supposed to add interior relief, but in this sample:

Land relief: 0.0341 -> 0.0329

This suggests the coast/strait/shelf components may be overpowering the intended interior relief.

Recommended Next Fixes

1. Reduce crust feedback

Separate crust cause fields from current height more strongly.

Current risky loop:

height / oceanDepthClass
-> crustThickness / crustAge / crustProvince
-> crust province terrain delta

-> height

Better:

plate + skeleton + tectonic context
-> crust causes
-> terrain influence

Height should be a weak modifier, not a primary classifier.

2. Add explicit diagnostics for cause-only reseed risk

Stages like these do not change height but can change export labels:

CRUST_FIELDS
FINAL_CONTINENT_RESEED
FINAL_CRUST_RESEED

Need a diagnostic like:

cause-only label changed export/color/debug risk

Because the current issue is not only terrain height. It is also final labels/materials/province metadata.

3. Treat CRUST_PROVINCE_DELTA as the main province terrain risk

This pass should probably have stronger limits based on:

province seam imprint
plate seam imprint
land-body count change
medium fragment change
export-height risk

Not only topology flips.

4. Keep crust skeleton but enforce that it remains repair-only

The current code says it is restrained, but the numbers show it still changes topology significantly.

It should maybe be measured by:

allowed repair improvement
vs
added imprint cost

If it reduces fragments but increases plate/province imprint too much, it should back off.

5. Rebalance quality pass for relief

If land relief is the goal, the current quality pass should be checked because on seed 67 it reduces land relief.

Possible causes:

coastal breakup overpowering interior relief
strait cuts reducing relief
shelf roughness not helping land relief
interior relief too weak or too gated

Final Diagnostic Conclusions

1. First plate/skeleton/province imprint jump:
   - Plate height imprint: SKELETON_ELEVATION
   - Skeleton identity imprint: CONTINENT_FIELDS
   - Province identity imprint: CRUST_FIELDS
   - Province terrain imprint: CRUST_PROVINCE_DELTA

2. Skeleton elevation over-amplifies plate-shaped structure before crust:
   - Yes, somewhat. It is the first major visible amplification stage.

3. Crust delta creates too many land bodies/fragments:
   - It adds land bodies, but not medium fragments in seed 67.

4. Crust cohere repairs crust delta damage:
   - Partly, but main repairs are crust skeleton obedience and tiny island cleanup.

5. Crust skeleton double-applies skeleton authority:
   - Yes. It is restrained but still a second skeleton terrain pass.

6. Crust fields create feedback:
   - Yes. Current crust fields are partly derived from height/oceanDepthClass, then used to modify height.

7. Stage that helps relief:
   - Best: CRUST_SKELETON_OBEDIENCE
   - Also helps slightly: SKELETON_ELEVATION
   - Hurts most: CRUST_PROVINCE_DELTA
   - Unexpectedly hurts: QUALITY_PASS

8. Pass that most increases export risk:
   - Terrain export risk: CRUST_PROVINCE_DELTA

- Metadata/debug/export label risk: CRUST_FIELDS
- Final label risk: FINAL_CRUST_RESEED